

Assignment 2 (Group)

Your assignment must be submitted to Moodle before the deadline. Create a folder and named it as **A2-GxTyy-UOWID1,UOWID2(Name1,name2)** where “x” stands for your tutorial group number and “yy” stands for the team number. For example, A2-G2T05 stands for team 05 in tutorial group 2. In this folder you should have one MS Words report and all your sources. **The sources can be in separate files.** Submit it via the Assignment Submission tab in moodle. Only 1 submission per group. **Part of the evaluation include the ability to follow these instructions.**

You need to demonstrate your work for this assignment. You may be asked questions during the demo about your code. Part of the evaluation include the ability to answer the question. You need to demo your work during the next tutorial after your submission. Team with **no demo will have their score cap at 50%.**

Assignment question: Using HTML, CSS, Javascript, XML, XSD, XSL and XLST to develop the content as shown below. There are 3 parts. Use relative link. All source codes need to follow proper programming style and convention to enhance its readability. Your report must include the screenshot of your webpage and the explanation of all your codes.

Part 1:

A table reflecting the following information. The format must be as shown below.

CSIT128: Assignment 2			Group GxTyy
Student Number / Name / Email	UOWID	Your Name	Your Email
	UOWID	Your Name	Your Email
	UOWID	Your Name	Your Email

Part 2:

This part is a game that the user can play. The game starts with the following screen in a table structure as shown below.

Part 2: Game		
Name:		
Your chosen character:	<list box showing B,F,\$,%,3,9>	Current Score: 0
<Start Game Button>		<Stop Game Button>
<1 st character>	<2 nd character >	<3 rd character >
Highest: 2 Tan Ah Dog		

The player will provide his/her name in the textbox and pick a target character from a list box showing “B”, “F”, “\$”, “%”, “3”, “9”. The initial value of score is zero and it must be shown as “0” in the box. The name is required, and all game features is disabled without a name.

When the player clicks on the “Start Game” button, in every 2 seconds, three random characters will be generated and displayed on the screen (e.g. 3 F \$) at the position as shown above. The three characters must not repeat at the same time (i.e. can’t have repeating characters – e.g. 3 F 3). After starting a game, you can’t change the chosen character.

To play the game, the player must click on the randomly generated character that was **chosen earlier to win 4 points**. If the player clicks on the wrong character, 3 points are deducted. Therefore, the game score can become negative if the player clicks on many wrong characters. For example, if the chosen character is “B” and the randomly generated characters are “B”, “3” and “\$”, clicking on the 1st character “B” will gain 4 points. Clicking on the character “3” will result in losing 3 points. The current score is displayed on the top right corner of the table. It must be updated in real-time as the player plays the game.

When the player clicks on the “Stop Game” button, the game is stopped. The screen must remain the same with the chosen character, the current score and the last 3 random characters unchanged. The game will maintain the highest score and the name of the player. The highest score and the name of the player will be displayed as shown above. When you click on “start” game again, it will reset the score to 0 and start the random character generation again. You can also change the chosen character to a new character. Try to have the font size, foreground and background colours as close as possible to what are shown above.

Part 3:

On Moodle site, under Assignment 2 section, download the file A2.xml and study it carefully. Using the content of A2.xml, create the following files.

- edited A2.xml file to use the stylesheet S2.xsl
- a S2.xsd file describing the structure of A2.xml.
 - You need to enforce the restriction for some of the xml data as follows:
 - <month> can be any number from 1 to 12.
 - <date> can be any number from 1 to 31.
 - <dayOfWeek> can be “Mon”, “Tues”, “Wed”, “Thur”, “Fri”, “Sat”, or “Sun”.
- a stylesheet file S2.xsl.

Below is only a sample showing the general format of the display. You should follow the given xml file and the formatting as described below. You should see different cells being filled up.

Singapore [01/08/20 10:00 PM]

Date	Mon	Tue	Wed	Thu	Fri	Sat	Sun
12 Jun					23° - 29°  Cloudy with a thunderstorm		
16 Jun		25° - 30°  Considerable clouds					
28 Jun							22° - 30°  A morning shower, then rain
06 Jul	30° - 34°  Plenty of sunshine						
10 Jul					29° - 33°  Partly sunny		
30 Jul				28° - 32°  Plenty of sunshine			

Please note that:

- The background colour for the first row and first column is **yellow**. The text for the 1st row is bold. All table cell contents are centralised.
- All the forecasts must be displayed in **descending** order from **latest date to earliest date**.
- The temperature is from **lowest to highest** with the degree symbol.
- You have to use the images from the A2 Resources folder (base on the “overallCode” element in the xml file)
- Use **blue** for “cloudy”, **orange** for “thunderstorm” and “rain” and **red** for “sunny” and “partly sunny” weather as shown above.